Roof Design Calculation

Given Data.:

Panel Width.: 4000 mm

Panel Length.: 2400 mm

Wind Load.: 14km/hr

PC Sheet Weight.: 2.5 kg

Intermediate Support tube.: 0.8 kg

A) <u>For Deflection Calculation</u>.:

1.Wind Speed.:

$V = 14{,}000/3600$

$= 3.89$ m/sec.

2.Wind Pressure (P).:

$\mathbf{P} = 0.613 \times V^2$

$= 0.613 \times (3.89)^2$

$= 0.613 \times 15.1321$

Wind Pressure (P) = 9.275 N/m^2 or Pa

3.Convert Given Load.:

Given load = PC Sheet Weight + Intermediate Support tube weight

$= 2.5 + 0.8$

$= 3.3$ kg per meter

Convert to 'N/m' –

= Mass x Accl^n due to gravity

$= 3.3 \times 9.81$

$= 32.37$ N/m

Convert to 'N/mm' –

$= 32.37 /1000$

Total Given Load = 0.03237 N/mm

Wind load per meter 'Width'

Area per meter Width = 1x4

Roof Design Calculation

$= 4m^2$

4.Wind Load (Wind force per meter).:

= P x A

= 9.275 x 4

= 37.1 N/m

Convert to 'N/mm' -

= 37.1/1000

= 0.0371 N/mm

Where:

P = 9.275 N/m^2

A = Width X Height

= 4 X 1

A = 4 m^2

5.Total Weight per 'mm'

Total Weight = Total Given Load + Wind load per meter

= 0.03237 + 0.0371

= 0.06947 N/mm

6.Section Properties.:

Width (B) = 63 mm

Height (H) = 60 mm

Thickness (t) = 2mm

b = (B-2t) = 63-2*2 = 59 mm

h = (H-2t) = 60-2*2 = 56 mm

7.Movement of Inertia (I).:

$I = (BH^3 – bh^3) / 12$

$I = ((63*60^3) - (59*56^3)) / 12$

I = 2,70,554 mm^4

Roof Design Calculation

8.Section Modulus (Z).:

$Z = I / Y$

Where:

$Y = H/2$

$= 60/2 = 30$ mm

$Z = 2,70,554 / 30$

$Z = 9018.48$ mm^3

9.Deflection Calculation.:

Simply Supported Beam.:

$= 5wL^4 / 384EI$

$= (5 \times 0.06947 \times 4000^4) / (384 \times 70,000 \times 2,70,554)$

Deflection =12.227 mm

10.Allowable deflection.:

For Roof:

Allowable = L/250

= 4000/250

= 16 mm

11.Final Result.:

Calculated deflection = 12.22mm

Allowable deflection = 16mm

Calculated deflection 12.22 mm is less than 16 mm allowable deflection

Since,

Design is safe in Deflection.

B) <u>For Bending Stress Calculation</u>

1.Maximum Bending Movement.:

$M = WL^2/8$

$= (0.06947 \times 4000^2)/8$

= 1,38,940 N-mm.

2.Section Modulus (Z) = 9018.48 mm^3

3.**Bending Stress ('N/mm').:**

Bending Stress = M/Z

= 1,38,940 / 9018.48

= 15.406 N/mm^2

= 15.406 M pa.

Allowable Stress = 0.6* fy

Or

Allowable Stress = fy/Factor of Safety

For Aluminum 6063-T4 fy = 110 MPA & Ultimate Strength = 150 MPA

1) Allowable stress = 0.6 x 110
   = 66 N/mm^2 allowable stress

2) Allowable stress = 110/1.65
   = 66.66 N/mm^2 allowable stress

   Where:

   For Aluminum FOS = 1.65 to 2

4.Final Result.:

15.406 M pa is less than 66 M pa allowable stress

Since**,**

Design is Safe in Bending Stress.